

DISCOVERY PHASE – COMPLETE (Zero assumptions; requirements locked verbatim from your query + full Sovereign Machina architecture + changelog v3.2.7 + latest repo state)

Core functionality required:

- Explore **GEPA profile evolution** — how the Genetic Prompt Evolution Algorithm (Phase 52) can dynamically evolve, refine, and adapt user profiles (daily-use, researcher, sovereign-red-game-master, etc.) over time within the current tri/quad-node stack.
- Focus on standalone personal intelligence use (voice-first assistant, knowledge OS, decision support, life logging, self-improving agents).
- Maintain: Node A = Mooncake KV buffer, Node B = Director, Node C = Oracle / Voice Core with Omi + Hermes, optional Hermes Pod. No changes to Node A or Node B core logic. Zero-trust, physical sovereignty, local-first design.

All requirements are now exhaustive and production-locked.

RESEARCH PHASE – RIGOROUS CROSS-CHECK (GEPA + Profile Mechanics)

GEPA (Genetic Prompt Evolution Algorithm) is a DSPy-based evolutionary system that maintains a population of prompt variants, scores them against high-signal trajectories from the Soul Logger, and evolves the fittest ones through crossover, mutation, and selection. It already refines system prompts, RAG queries, and shard configurations.

User profiles are defined in SOVEREIGN-IDENTITY.md as named override sets (tone, memory_weight, tools, response_style, ui_theme). These are loaded by Hermes and influence enrichment, tool access, and response formatting.

The combination is natural: GEPA can treat each profile's tokens as evolvable parameters, using real usage data (voice notes, screen captures, user feedback, outcome quality) as the fitness function.

EXPLORATION OF GEPA PROFILE EVOLUTION

1. How GEPA Profile Evolution Works in the Current Stack

Core Mechanism

GEPA runs as a background service on the Hermes Pod (or Node C in tri-node mode). It treats each profile as an evolvable “genome”:

- **Genome:** The set of profile tokens (tone, memory_weight, tools, response_style, etc.).

- **Population:** Multiple variants of each profile (e.g., 8–12 variants of “researcher”).
- **Fitness Function:** Scored on real usage data:
 - User satisfaction (explicit thumbs-up/down or implicit continuation rate).
 - Outcome quality (Ouroboros audit scores, RKG consistency).
 - Efficiency (latency, token usage, resource consumption).
 - Relevance (how well the profile matches the intent context).

Evolution Cycle (Run on Hermes Pod)

1. **Input Collection:** Every voice note, screen capture, or interaction is tagged with the active profile and logged with provenance.
2. **Trajectory Scoring:** Soul Logger + Ouroboros scores each interaction against the active profile variant.
3. **Genetic Operations:**
 - **Crossover:** Combine high-scoring tokens from different variants (e.g., researcher tone + daily-use memory_weight).
 - **Mutation:** Small random edits (e.g., adjust memory_weight from “deep historical” to “deep historical + recent”).
 - **Selection:** Keep top-N variants (elitism + diversity).
4. **Atomic Promotion:** The winning variant becomes the new active profile for that context. Changes are written to SOVEREIGN-IDENTITY.md (git commit) and loaded instantly by Hermes.
5. **Ouroboros Gate:** Every evolved profile is audited before promotion to ensure it does not violate global invariants.

Frequency: Run every 5–15 minutes during active use, or as a nightly “Dream Cycle” batch for deeper evolution.

2. Concrete Benefits for Intelligence Boost

- **Contextual Adaptation:** The “researcher” profile might evolve to weight recent papers higher if you frequently ask for cutting-edge research, while the “daily-use” profile learns your personal shorthand and preferences.
- **Self-Improvement Loop:** Profiles get smarter over time. GEPA uses real outcome data to refine how the assistant behaves in each context.
- **Reduced Cognitive Load:** The system automatically adapts to your changing needs without manual reconfiguration.
- **Cross-Profile Learning:** High-signal tokens from one profile can migrate to others (e.g., a useful `response_style` from researcher mode improves daily-use mode).
- **Auditability:** Every profile evolution is logged immutably with provenance, feeding back into Ouroboros for system-wide self-healing.

3. Integration with Existing Components

- **Hermes Supervisor:** Owns the GEPA evolution loop and applies the active profile to every intent.
- **GBrain / MemPalace:** Stores profile-specific pages and relationships (e.g., “evolved researcher profile linked to 47 GEPA notes”).
- **Node C (Oracle):** Voice notes and screen captures are tagged with the active profile for better enrichment.
- **Node B (Director):** Receives profile-scoped context for higher-quality, context-appropriate reasoning.
- **Machina Terminal:** Displays current profile in the header and adapts UI subtly (e.g., researcher mode shows more citations).
- **Ouroboros:** Audits every evolution for consistency with global invariants.

4. Practical Example (Daily Use Case)

Morning voice note: “Switch to researcher mode and summarize the latest papers on GEPA culling.”

- Hermes loads the current “researcher” profile.

- GEPA has already evolved it based on last week's usage (increased weight on recent sources, added "include counterarguments" rule).
- Node C (Omi) transcribes, Hermes enriches with RKG + profile tokens.
- Node B (Director) produces a structured report with citations and links to your existing notes.
- User feedback ("good summary") is logged and used in the next GEPA cycle to further refine the profile.

ARCHITECT SIGN-OFF

GEPA profile evolution is a natural and powerful extension of the persistent identity logic. By treating each profile as an evolvable genome scored on real usage trajectories, GEPA turns static profiles into living, adaptive contexts that improve over time. This deepens the self-improving nature of Sovereign Machina without touching Node A or Node B core logic. The mechanism fits cleanly into the Hermes supervisor on the Hermes Pod (or Node C) and leverages existing components (Soul Logger, Ouroboros, MemPalace) for maximum intelligence boost while preserving full sovereignty and auditability.

Your main sovereign machine remains completely untouched and rock-solid.

Clarifying questions (mandatory before next phase):

1. Do you want the starter SOVEREIGN-IDENTITY.md template with GEPA-evolvable profile sections?
2. Shall I include the Nix/Hermes code for the GEPA profile evolution loop?

Reply with your priority and I will deliver the concrete implementation. The GEPA profile evolution is ready to be prototyped.